# FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

**Syllabus: Ch 1 — Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1–7.6)**

**Total Marks: 65** **Time: 2h 30m** **Version: 5**

**General Instructions:**
1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

## SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

| # | Question | A | B | C | D | Answer |
|---|---|---|---|---|---|---|
| 1 | “Biodiversity” refers to: | ○Variety of habitats only | ○Variety of living organisms on Earth | ○Number of endangered species | ○ Classification system for animals | |
| 2 | Which is NOT listed as an ecological/economic importance of biodiversity? | ○Oxygen and clean water | ○Pollination of crops | ○Formation of fossil fuels | ○Medicines and industrial materials | |
| 3 | Which is NOT one of the characteristics shared by all living organisms? | ○Movement | ○Sensitivity | ○ Crystallization and storage | ○Growth | |
| 4 | The scientific study of diversity of organisms and their evolutionary relationship is called: | ○Taxonomy | ○Nomenclature | ○Systematics | ○Morphology | |
| 5 | Which statement about the cytoplasm of a cell is correct? | ○90% water; site of biochemical reactions | ○10% water; contains no organelles | ○Found only in plant cells | ○Same structure as nucleoplasm | |
| 6 | Each higher level of organization (e.g. tissue) gaining a functionality absent at the level below (e.g. cell) is called: | ○Homeostasis | ○Differentiation | ○Emergent properties | ○ Specialization | |
| 7 | Benedict's test, giving a green/yellow/red precipitate on boiling, detects the presence of: | ○Starch | ○Proteins | ○Reducing sugars | ○Lipids | |
| 8 | Which protein is a major component of skin, bones, cartilage, tendons and ligaments (support/strength)? | ○Keratin | ○Collagen | ○Haemoglobin | ○Fibrinogen | |
| 9 | Actin and myosin, present in muscle cells, are mainly responsible for: | ○Blood clotting | ○Muscle contraction/relaxation | ○Oxygen transport in blood | ○Antibody formation | |
| 10 | A few enzymes are RNA in nature, not protein. These are called: | ○Isozymes | ○Coenzymes | ○Ribozymes | ○Apoenzymes | |
| 11 | Studying production of wheat, fish and rice and their export value is an example of: | ○Biostatistics | ○Bio-economics | ○ Biogeography | ○ Biotechnology | |
| 12 | Unlike Interphase I, Interphase II (between Meiosis I and II): | ○Involves full DNA replication | ○Involves no further DNA replication | ○Involves crossing over | ○Involves synapsis of chromosomes | |

## SECTION B — Short Questions (11 × 3 = 33 Marks)

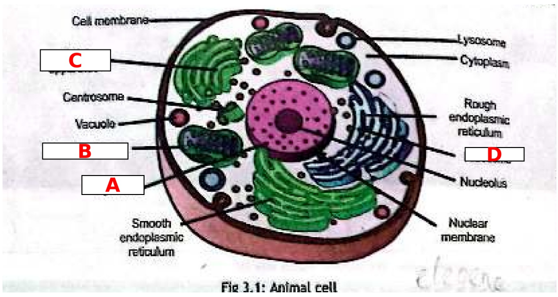
Attempt 11 questions. Choose either the main question OR its alternative.

| Part | Question | Marks | OR | Alternative Question | Marks |
|---|---|---|---|---|---|
| 2(i) | Define cytoplasm. State its approximate water content and TWO functions. | 3 | OR | What is the significance of liver cells being described as the ‘most metabolically active’ cells in the body? State any TWO functions of liver cells. | 3 |
| 2(ii) | List the SEVEN characteristics shared by all living | 3 | OR | Explain the concept of ‘division of labour’ within a | 3 |

| | | | | | |
|---|---|---|---|---|---|
| **2(ii)** | organisms, as introduced in the chapter on biodiversity. | **3** | **OR** | cell, and how this idea extends to cells working together in a multicellular organism. | **3** |

**2(iii)** **3 marks**

The figure below shows an animal cell (Fig. 3.1, adapted from your textbook). Identify the structures labelled A, B, C, and D.




A = ____________________ B = ____________________ C = ____________________ D = ____________________

**OR** Describe the structure of the nucleolus and explain what happens to it during different stages of cell division. **[3 marks]**

| Part | Question | Marks | OR | Alternative Question | Marks |
|---|---|---|---|---|---|
| **2(iv)** | Describe the Benedict's test and the Fehling's test used to detect reducing sugars such as glucose, including the colour changes observed. | **3** | **OR** | Name any FIVE functions of proteins in the human body, giving ONE specific example protein for each function. | **3** |
| **2(v)** | What are ribozymes? How are they different from the majority of enzymes, which are globular proteins? | **3** | **OR** | Explain why enzymes are required only in very small quantities within a cell and can be used repeatedly for the same reaction. | **3** |
| **2(vi)** | Explain the concept of 'emergent properties' in the levels of biological organization, giving ONE example. | **3** | **OR** | Describe the structure and function of the upper and lower epidermis of a leaf, including the role of the cuticle. | **3** |
| **2(vii)** | State the importance of biochemistry in studying living organisms, according to the textbook (any TWO points). | **3** | **OR** | Differentiate between a monomer and a polymer. Use glucose and starch as examples in your answer. | **3** |
| **2(viii)** | State the observations made by A.F.A. King in 1883 regarding the spread of malaria. What hypothesis did these observations lead him to form? | **3** | **OR** | Differentiate between qualitative and quantitative observations in the biological method, giving ONE example of each. | **3** |
| **2(ix)** | Name TWO plant sources and TWO animal sources of proteins, as mentioned in the textbook. | **3** | **OR** | What is a wax? Name ONE natural source of wax mentioned in the textbook and state ONE of its functions. | **3** |
| **2(x)** | According to cell theory, new cells arise by division of a pre-existing cell. Name any THREE reasons why cell division is necessary in living organisms. | **3** | **OR** | What is the mitotic (M) phase of the cell cycle? Name its TWO major sub-processes. | **3** |
| **2(xi)** | Define biodiversity. State any THREE ways it is ecologically or economically important to human life, as described in the textbook. | **3** | **OR** | What is systematics? How is it different from taxonomy? | **3** |

## SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

**3(i)** **5 marks**

Draw a well-labelled diagram of the transverse section of a leaf, showing at least SIX parts (e.g. upper epidermis, palisade mesophyll, spongy mesophyll, xylem, phloem, guard cells and stoma).

*(space for diagram)*

**OR** Explain how the internal structure of a leaf is adapted for photosynthesis and transpiration. Refer to the cuticle, the palisade and spongy mesophyll, and the arrangement of stomata in your answer. **[5 marks]**

| Part | Question | Marks | OR | Alternative Question | Marks |
|---|---|---|---|---|---|
| **3(ii)** | (a) Define biodiversity. Discuss its ecological and economic importance to humans under any FOUR headings, as described in the textbook. [3]<br>(b) What is systematics, and how does it differ from classification? [2] | **5** | **OR** | (a) List the SEVEN characteristics common to all living organisms, as introduced in the biodiversity chapter. [3]<br>(b) Name the EIGHT taxonomic ranks in the correct order, from Domain to Species, used in the modern classification hierarchy. [2] | **5** |
| **3(iii)** | Describe the different functions performed by proteins in the human body. Explain in detail the roles of: (a) collagen and keratin as structural proteins, (b) actin and myosin in muscle contraction, (c) haemoglobin in oxygen transport, and (d) antibodies in immunity. | **5** | **OR** | Describe THREE tests used to identify biomolecules in food samples: (a) the iodine test for starch, (b) the Benedict's/Fehling's test for reducing sugars, and (c) the Biuret test for proteins. State the reagent/procedure and the resulting colour change for each. | **5** |
| **3(iv)** | (a) Name the FOUR fat-soluble vitamins mentioned in the textbook. Which one is synthesized from cholesterol in the body? [2]<br>(b) Steroids are a distinct class of lipids. Describe their structure and name TWO examples mentioned in the textbook. [3] | **5** | **OR** | (a) Differentiate between a tri-acyl glyceride and a phospholipid in terms of structure (number of fatty acid chains attached to glycerol). [2]<br>(b) Lipids contain more than double the energy of carbohydrates per gram. Explain this fact in terms of their chemical composition (carbon-hydrogen ratio). [3] | **5** |

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

Chapters Covered: Ch 1 — Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1–7.6)

**— End of Question Paper —**